

Phase I/II Operating Characteristics

Non-TITE alpha=0 (IPDE-first) versus U-BOIN, BOIN12, and EffTox

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
Non-TITE C2 one-stage	8.3	6.3	2.6	1.6	0.2	81
Non-TITE C3 one-stage	9.4	6.7	3.3	1.3	0.2	79.1
Non-TITE C2 two-stage	8.5	6.1	2.7	1.6	0.2	80.9
Non-TITE C3 two-stage	9.8	7.4	3.2	1.2	0.3	78.1
U-BOIN	15.6	17.4	20.6	18.7	13.7	14
BOIN12	17.5	12	4.8	1.4	0.9	63.4
EffTox	5	4	6	3	6	75
Mean administrations by dose						
Non-TITE C2 one-stage	10.6	4.7	2.7	1.1	0.4	
Non-TITE C3 one-stage	10.8	4.7	2.6	1.1	0.4	
Non-TITE C2 two-stage	10.5	4.8	2.8	1.2	0.4	
Non-TITE C3 two-stage	10.8	4.8	2.6	1.1	0.4	
U-BOIN	9	5.4	4.7	3.9	3.7	
BOIN12	10	5.8	3.3	1.9	0.8	
EffTox	8	3.2	3	2	2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
Non-TITE C2 one-stage	22.4	22.4	19.9	18	15.8	1.5
Non-TITE C3 one-stage	23.9	22.1	18.3	19	15.7	1
Non-TITE C2 two-stage	22	22	19.4	18.9	16.3	1.4
Non-TITE C3 two-stage	23.1	22.5	18.5	19.7	15.2	1
U-BOIN	28.1	26.8	21.1	16.3	7.7	0
BOIN12	35.1	27.5	19.7	12.1	5.6	0
EffTox	66	5	8	7	9	5
Mean administrations by dose						
Non-TITE C2 one-stage	7.1	6.5	5.9	5.5	5	
Non-TITE C3 one-stage	7.1	6.4	5.7	5.9	4.9	
Non-TITE C2 two-stage	6.9	6.4	6	5.7	5.1	
Non-TITE C3 two-stage	6.9	6.4	5.7	6	5.1	
U-BOIN	6.1	6.2	6.4	6.4	5.4	
BOIN12	4.9	5.7	6.7	7.4	5.3	
EffTox	20.9	2.3	2.5	1.7	2.1	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
Non-TITE C2 one-stage	4.1	42.8	31.7	12.7	3.1	5.6
Non-TITE C3 one-stage	4	46.3	28.6	13.7	2.4	5
Non-TITE C2 two-stage	5.1	43.8	30.8	12.5	2.8	5
Non-TITE C3 two-stage	4.2	45.4	29.9	12.9	2.3	5.3
U-BOIN	8.5	47.5	18.6	12.4	13	0
BOIN12	11.3	62.1	18.8	4.9	0.1	2.8
EffTox	24	30	20	7	10	9
Mean administrations by dose						
Non-TITE C2 one-stage	5.2	10.6	7.9	4.3	1.6	
Non-TITE C3 one-stage	5.2	10.6	7.7	4.4	1.7	
Non-TITE C2 two-stage	5	10.1	8	4.7	1.9	
Non-TITE C3 two-stage	5.1	10.3	7.9	4.5	1.9	
U-BOIN	5.9	10.8	7	4	2.4	
BOIN12	6.2	10.7	7	4.1	1.7	
EffTox	13.5	6.6	5.1	2	1.7	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
Non-TITE C2 one-stage	1	1.3	4.7	59.8	33.2	0
Non-TITE C3 one-stage	0.5	1.8	7	57.3	33.4	0
Non-TITE C2 two-stage	0.8	1.1	5.7	55.8	36.5	0.1
Non-TITE C3 two-stage	0.3	2.1	5.4	58.3	33.9	0
U-BOIN	1.5	3.2	8.2	60.1	27	0
BOIN12	1.7	3.3	6.7	65.7	22.6	0
EffTox	5	5	10	18	63	1
Mean administrations by dose						
Non-TITE C2 one-stage	3.5	3.4	4.5	12.2	6.4	
Non-TITE C3 one-stage	3.4	3.5	4.5	12	6.6	
Non-TITE C2 two-stage	3.4	3.4	4.6	10.6	8	
Non-TITE C3 two-stage	3.4	3.5	4.3	11.2	7.7	
U-BOIN	3.5	4	5.1	9.8	7.9	
BOIN12	3.2	3.4	4.7	12.9	5.8	
EffTox	8.1	3.3	4.6	4	9.9	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
Non-TITE C2 one-stage	11.3	21.6	49.3	11	0.7	6.1
Non-TITE C3 one-stage	10.7	23.5	46.5	11.7	1.3	6.3
Non-TITE C2 two-stage	10.7	23.9	48.7	9.9	0.8	6
Non-TITE C3 two-stage	10.1	23.5	48	11.1	0.8	6.5
U-BOIN	8.7	20.9	38.9	16.5	14.5	0.5
BOIN12	20.9	32.9	38	4.2	0	4
EffTox	36	27	15	12	6	4
Mean administrations by dose						
Non-TITE C2 one-stage	7.1	7.8	10.3	3.5	0.6	
Non-TITE C3 one-stage	7	7.8	9.9	3.9	0.6	
Non-TITE C2 two-stage	6.8	7.7	9.9	4	1.1	
Non-TITE C3 two-stage	6.7	7.6	9.7	4.2	1	
U-BOIN	6.9	8.1	8.8	4.3	1.8	
BOIN12	7.2	9	9.2	3.4	0.7	
EffTox	16.6	5.1	3.7	2.2	1.7	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
Non-TITE C2 one-stage	4.9	5.7	8.9	11.8	64.5	4.2
Non-TITE C3 one-stage	4.5	5.7	8.8	12.3	65	3.7
Non-TITE C2 two-stage	4.8	5.9	10	11.5	63.7	4.1
Non-TITE C3 two-stage	4.9	5.8	7.7	12.4	65.6	3.6
U-BOIN	6.2	8.9	13	13.3	58.6	0
BOIN12	13.2	13.8	16.5	13.6	41.5	1.4
EffTox	1	2	8	15	73	2
Mean administrations by dose						
Non-TITE C2 one-stage	4.7	4.3	5.3	5.4	10.2	
Non-TITE C3 one-stage	4.7	4.5	5	5.2	10.6	
Non-TITE C2 two-stage	4.7	4.3	5.2	5.4	10.4	
Non-TITE C3 two-stage	4.6	4.3	4.9	5.4	10.7	
U-BOIN	5.2	5.5	5.6	6.3	7.8	
BOIN12	4.2	4.7	5.2	6.1	9.6	
EffTox	3.4	3.3	4.5	5.2	13.3	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
Non-TITE C2 one-stage	12.7	18.2	19	19.5	19.5	18.9	17.9
Non-TITE C3 one-stage	10.6	14.4	15.7	16.5	16.5	15.5	14.9
Non-TITE C2 two-stage	12.7	18.2	19.1	19.6	19.7	18.9	18
Non-TITE C3 two-stage	10.7	14.5	15.8	16.7	16.8	15.6	15
U-BOIN	26.7	30.5	30.1	30.3	29.9	30.4	29.7
BOIN12	21.8	30	29.7	30	29.5	29.8	28.5
EffTox	18.2	29.5	28.9	29.9	29.3	29.7	27.6

Values are reported from Total_Unique_Patients; comparator counts without patient IDs are sums of dose-level means.

No OBD is source-specific: non-TITE=No OBD; U-BOIN=Stop; BOIN12=No selection; EffTox=None selected.